Chapter 2: Atomic Structure & Interatomic Bonding

ISSUES TO ADDRESS...

- What promotes bonding?
- What types of bonds are there?
- What properties are inferred from bonding?

Atomic Structure (Freshman Chem.)

- atom – electrons – 9.11×10^{-31} kg
  protons, neutrons } 1.67×10^{-27} kg

- atomic number can be defined as the number:

1. of protons in nucleus of atom.
2. of electrons in neutral species.

- A [=] atomic mass unit = amu = 1/12 mass of ^{12}C

Atomic wt = wt of 6.022×10^{23} molecules or atoms

1 amu/atom = 1 g/mol

C	12.011
H	1.008 etc.

Atomic Structure

- Some of the following properties
  1) Chemical
  2) Electrical
  3) Thermal
  4) Optical

  are determined by electronic structure

Electronic Structure

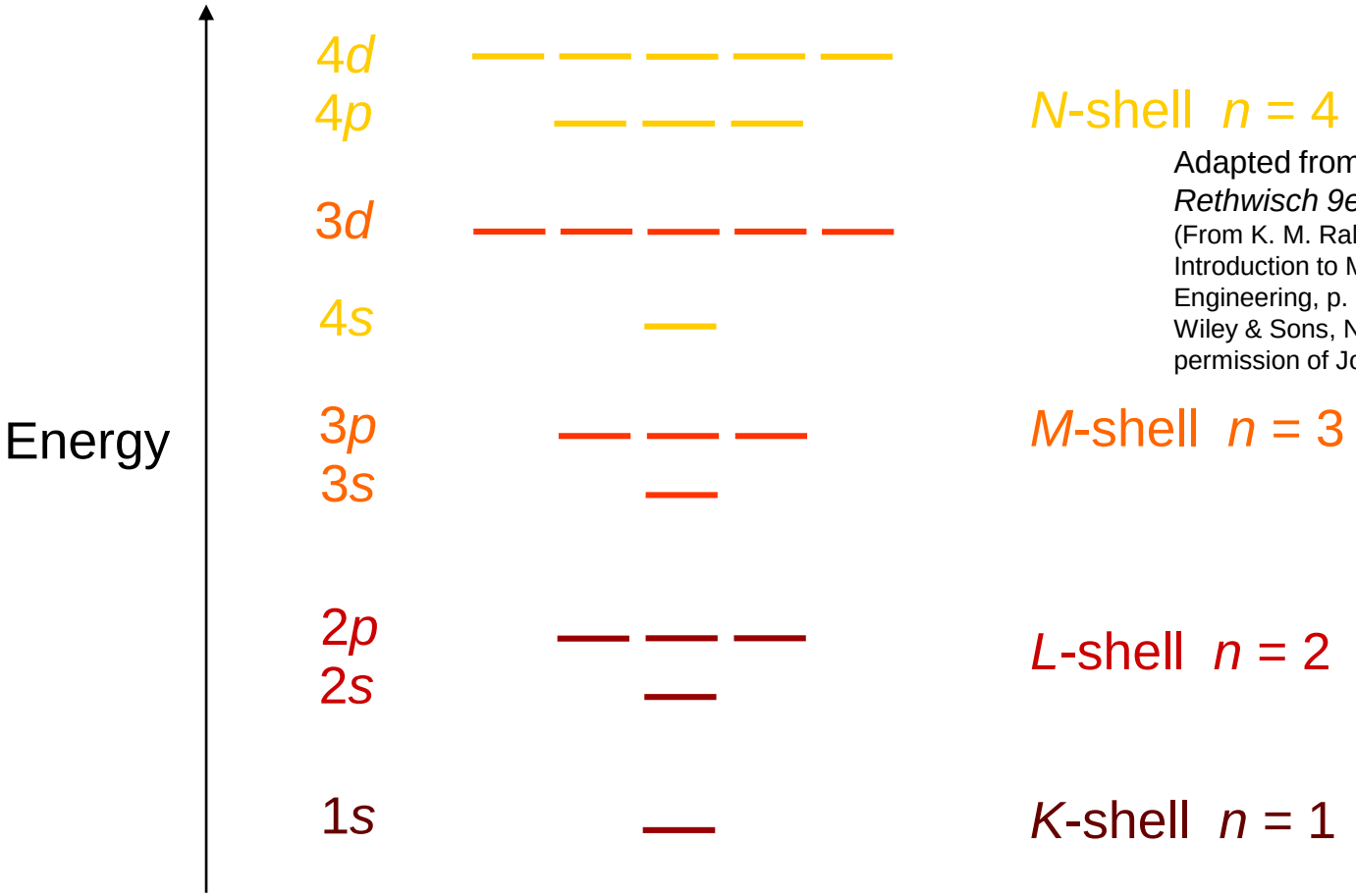
- Electrons have wavelike and particulate properties.
- Two of the wavelike characteristics are
  - electrons are in orbitals defined by a probability.
  - each orbital at discrete energy level is determined by quantum numbers.

– Quantum Number	Designation
n = principal (energy level-shell)	K, L, M, N, O (1, 2, 3, etc.)
ℓ = subsidiary (orbitals)	s, p, d, f (0, 1, 2, 3,…, n-1)
m_l = magnetic	1, 3, 5, 7 (-ℓ to +ℓ)
m_s = spin	½, -½

Electron Energy States

Electrons...

- have discrete energy states
- tend to occupy lowest available energy state.



Adapted from Fig. 2.6, *Callister & Rethwisch 9e.*
(From K. M. Ralls, T. H. Courtney, and J. Wulff, Introduction to Materials Science and Engineering, p. 22. Copyright © 1976 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.)

SURVEY OF ELEMENTS

- Most elements: Electron configuration not stable.

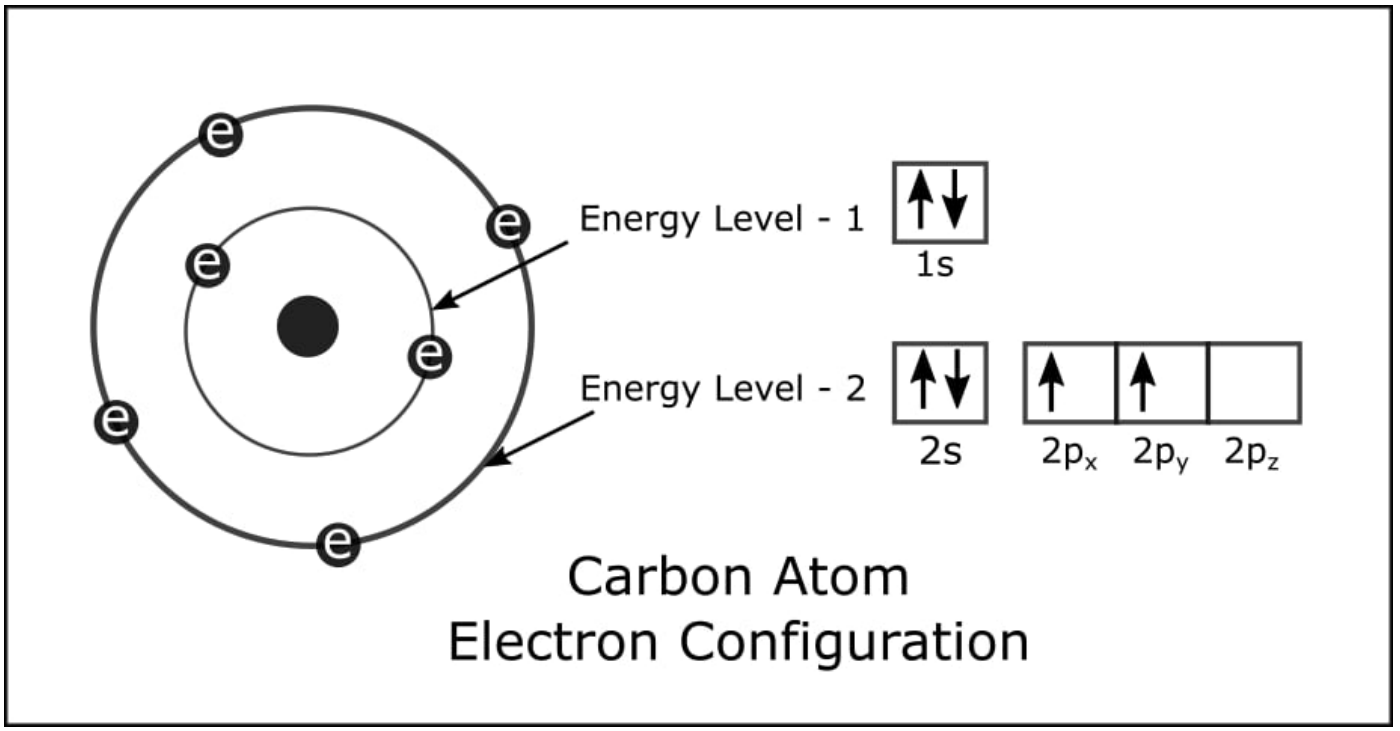
Element	Atomic #	Electron configuration	
Hydrogen	1	$1s^1$	
Helium	2	$1s^2$	(stable)
Lithium	3	$1s^2 2s^1$	
Beryllium	4	$1s^2 2s^2$	
Boron	5	$1s^2 2s^2 2p^1$	
Carbon	6	$1s^2 2s^2 2p^2$	
...		...	
Neon	10	$1s^2 2s^2 2p^6$	(stable)
Sodium	11	$1s^2 2s^2 2p^6 3s^1$	
Magnesium	12	$1s^2 2s^2 2p^6 3s^2$	
Aluminum	13	$1s^2 2s^2 2p^6 3s^2 3p^1$	
...		...	
Argon	18	$1s^2 2s^2 2p^6 3s^2 3p^6$	(stable)
...	...	...	
Krypton	36	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$	(stable)

Adapted from Table 2.2, *Callister & Rethwisch 9e.*

- Why? Valence (outer) shell usually not filled completely.

Electron Configurations

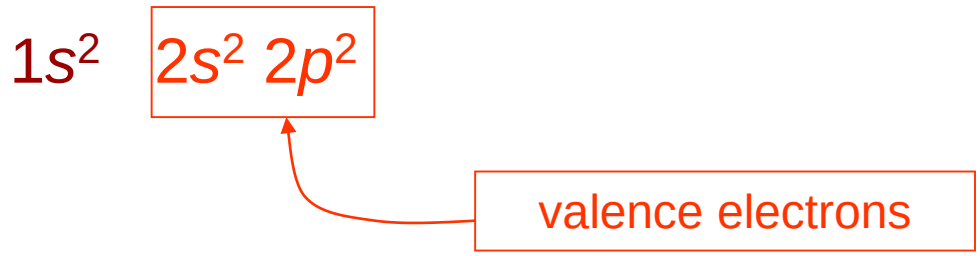
The outermost shell of any atom is called the valence shell and the electrons that reside in the valence shell are called valence electrons.



Electron Configurations

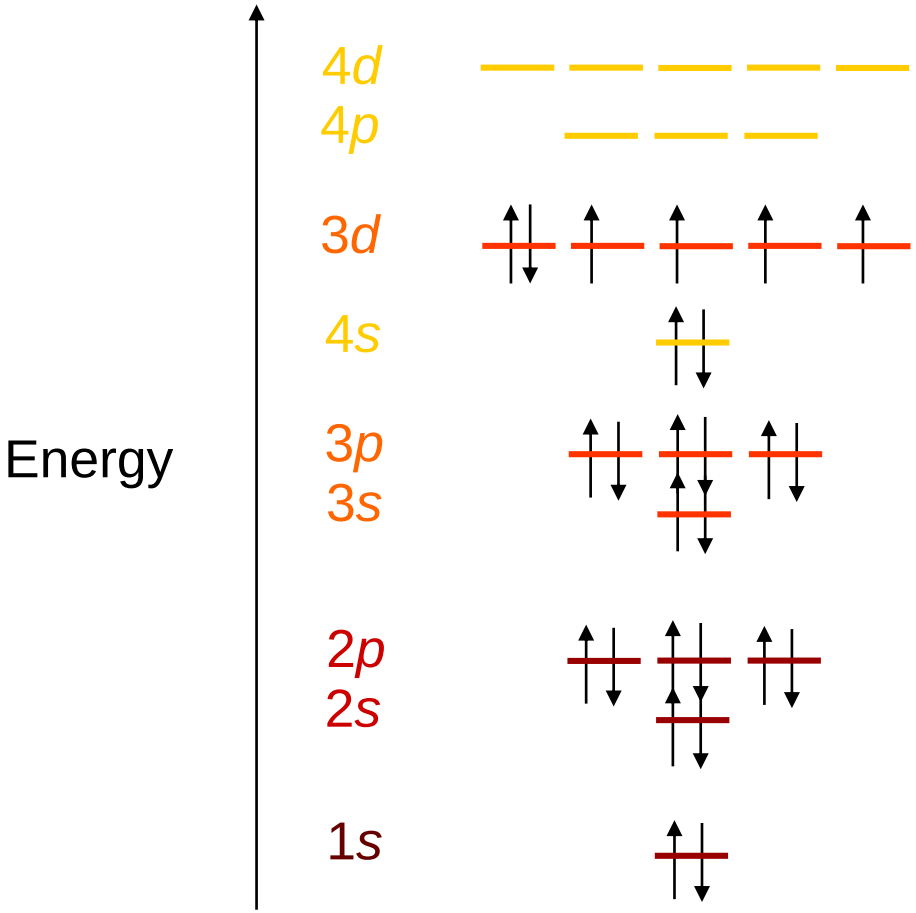
- Valence electrons – those in unfilled shells
- Filled shells more stable
- Valence electrons are most available for bonding and tend to control the chemical properties

  – example: C (atomic number = 6)



Electronic Configurations

ex: Fe - atomic # = 26 $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $3d^6$ $4s^2$



N-shell $n = 4$

valence electrons

M-shell $n = 3$

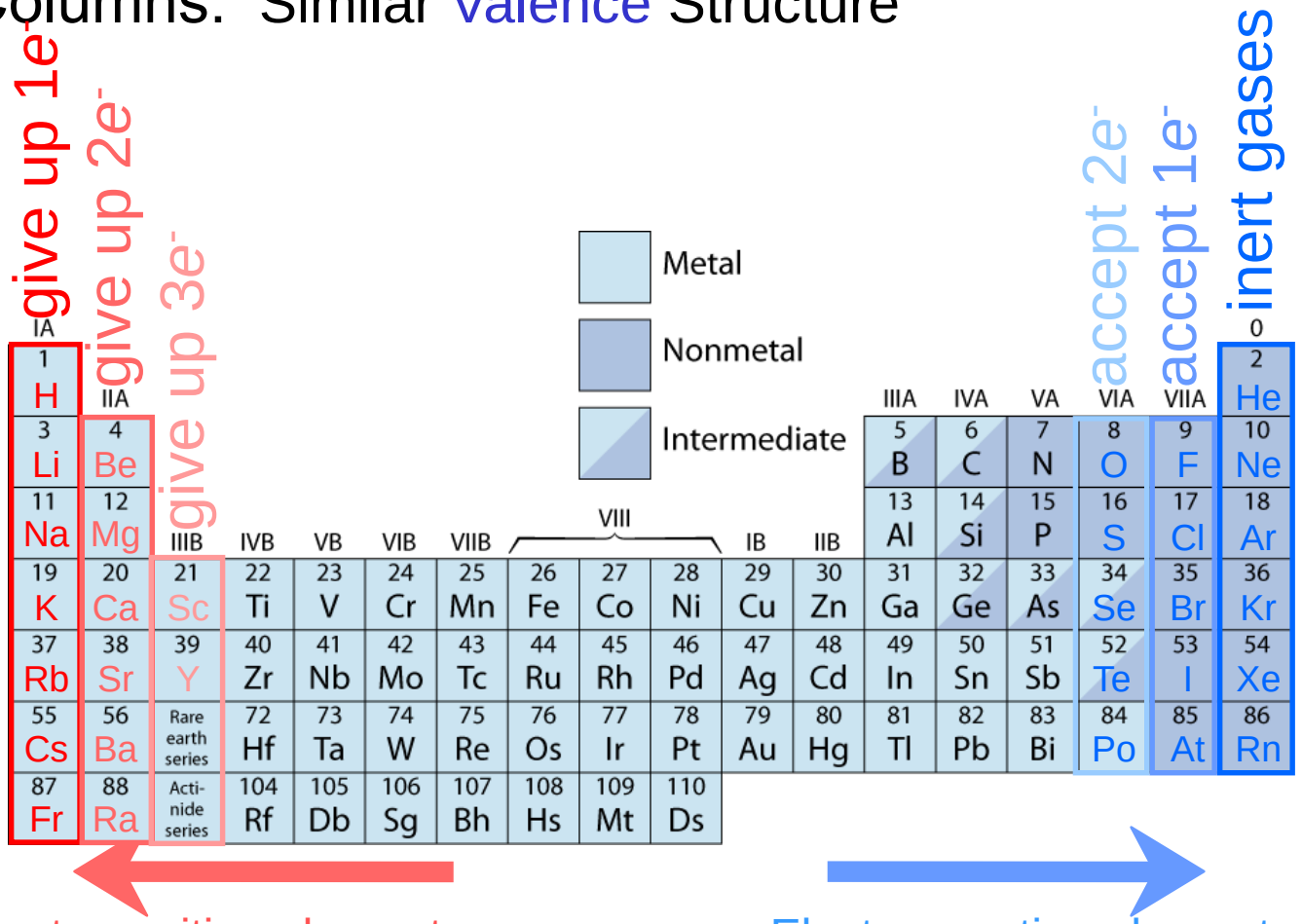
L-shell $n = 2$

K-shell $n = 1$

Adapted from Fig. 2.6, *Callister & Rethwisch 9e.*
(From K. M. Ralls, T. H. Courtney, and J. Wulff, Introduction to Materials Science and Engineering, p. 22. Copyright © 1976 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.)

The Periodic Table

- Columns: Similar Valence Structure



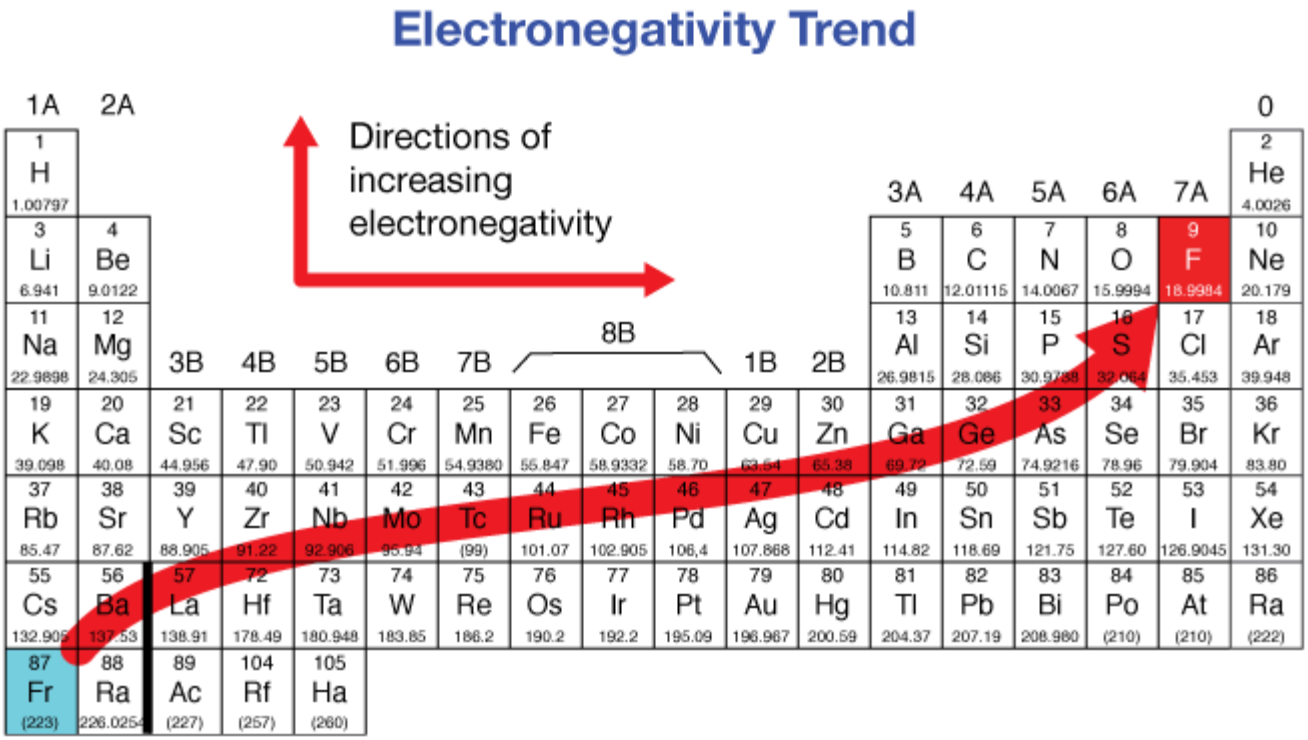
Adapted from Fig. 2.8, *Callister & Rethwisch 9e.*

Electropositive elements: Readily give up electrons to become + ions.

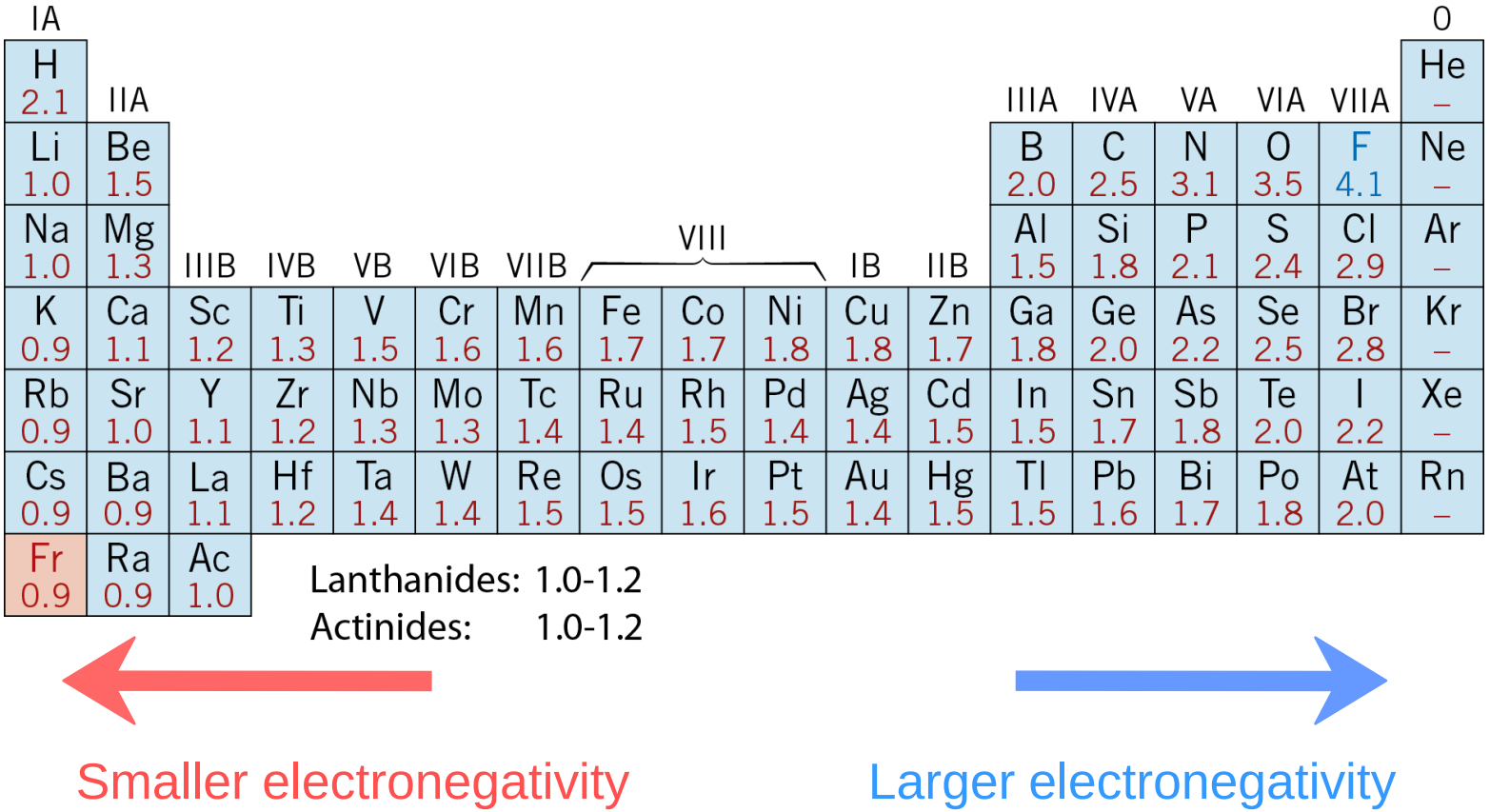
Electronegative elements: Readily acquire electrons to become - ions.

Electronegativity

Electronegativity is **a chemical property that describes the tendency of an atom or a functional group to attract electrons toward itself**.

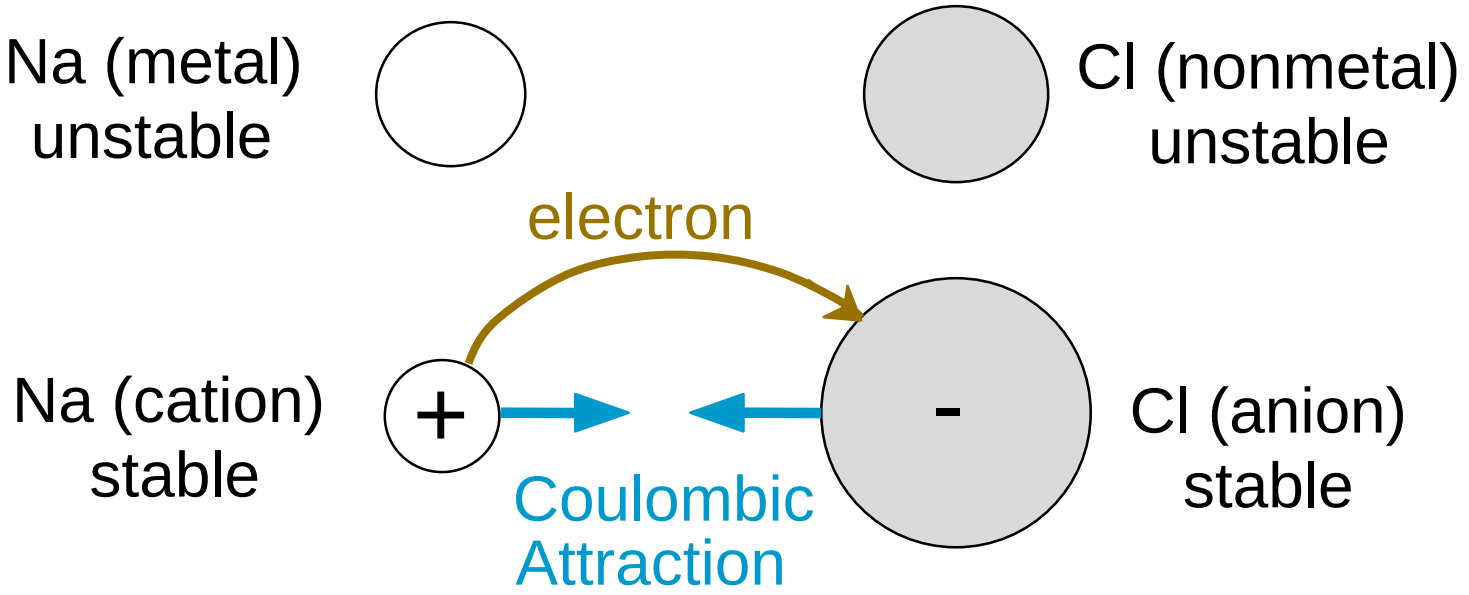


Electronegativity



Ionic Bonding

- Occurs between + and - ions.
- Requires electron transfer.
- Large difference in electronegativity required.
- Example: NaCl



Ionic bond is a bond that if formed between a metal + nonmetal

donates electrons

accepts electrons

Dissimilar electronegativities

ex: MgO

Mg $1s^2\ 2s^2\ 2p^6\ 3s^2$ or [Ne] $3s^2$

O $1s^2\ 2s^2\ 2p^4$

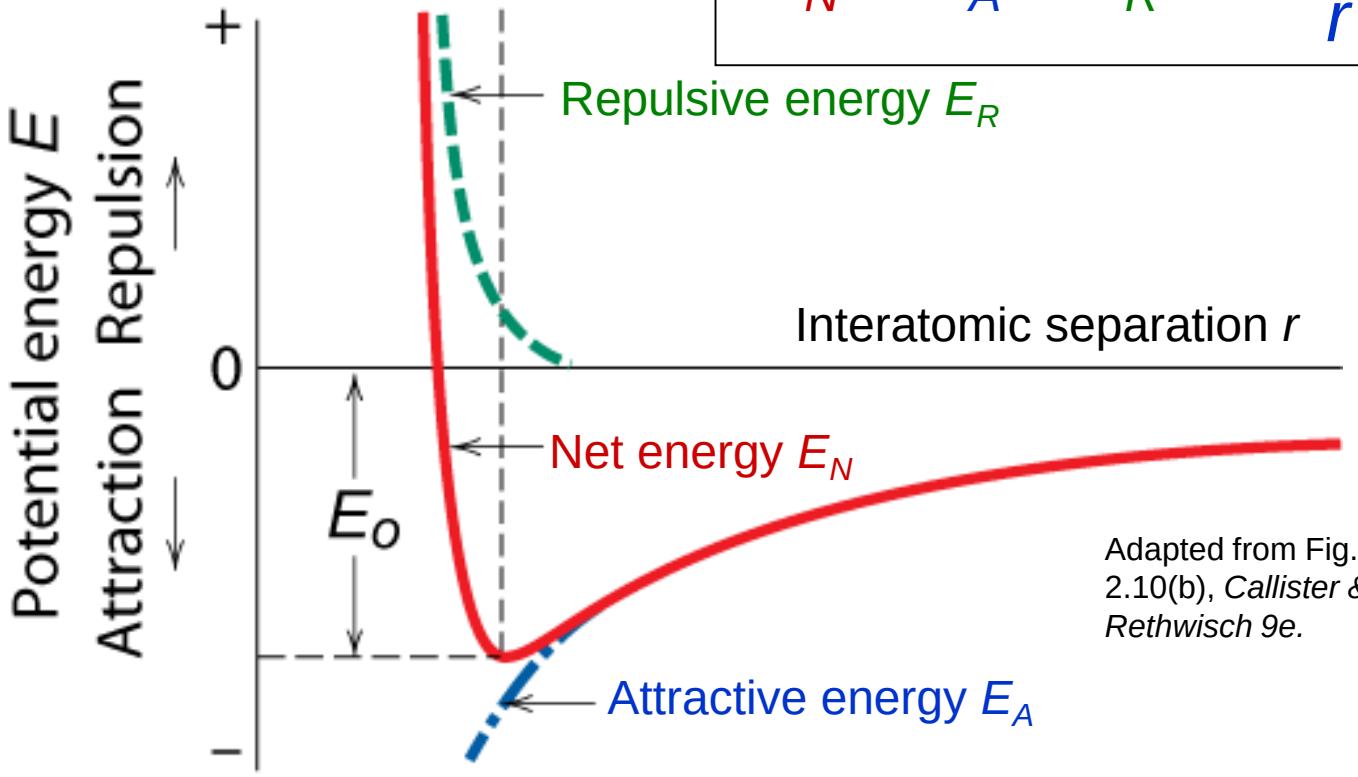
Mg^{2+} $1s^2\ 2s^2\ 2p^6$ [Ne]

O^{2-} $1s^2\ 2s^2\ 2p^6$ [Ne]

Ionic Bonding

- Energy – minimum energy most stable
  - Energy balance of attractive and repulsive terms

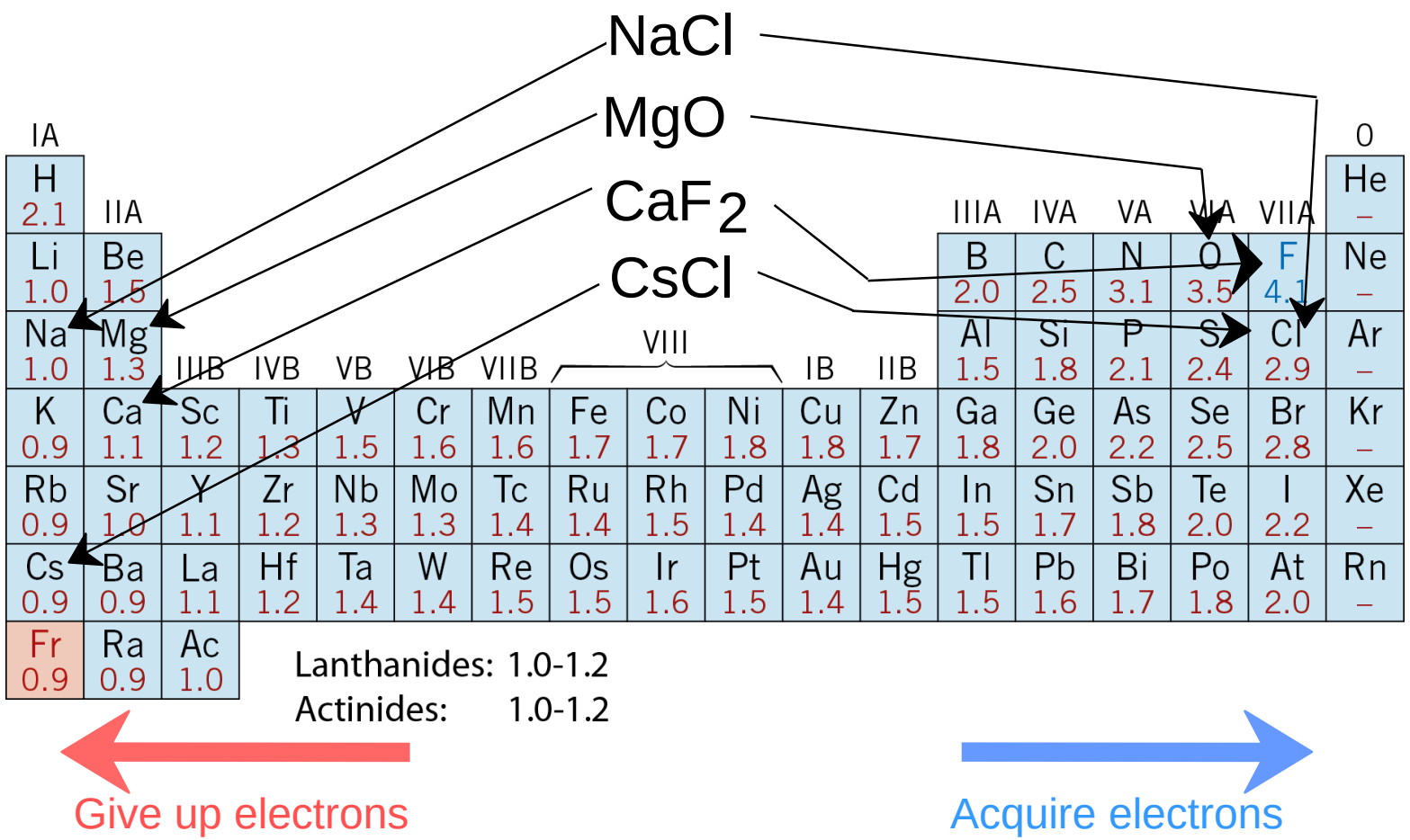
$$E_N = E_A + E_R = -\frac{A}{r} + \frac{B}{r^n}$$



Adapted from Fig. 2.10(b), *Callister & Rethwisch 9e.*

Examples: Ionic Bonding

- Predominant bonding in Ceramics



Covalent Bonding

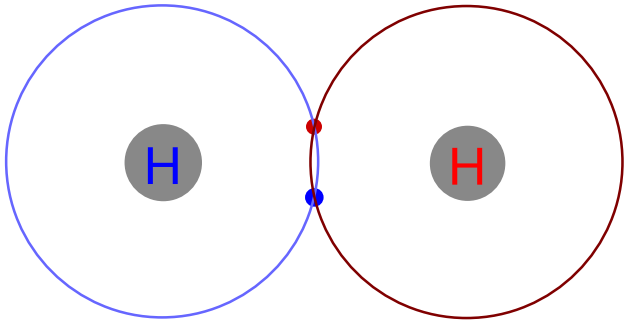
A covalent bond *is formed by the equal sharing of electrons from both participating atoms*.

- similar electronegativity ∴ share electrons
- bonds determined by valence – *s* & *p* orbitals dominate bonding
- Example: H_2

Each H: has 1 valence e^-, needs 1 more

Electronegativities are the same.

H_2

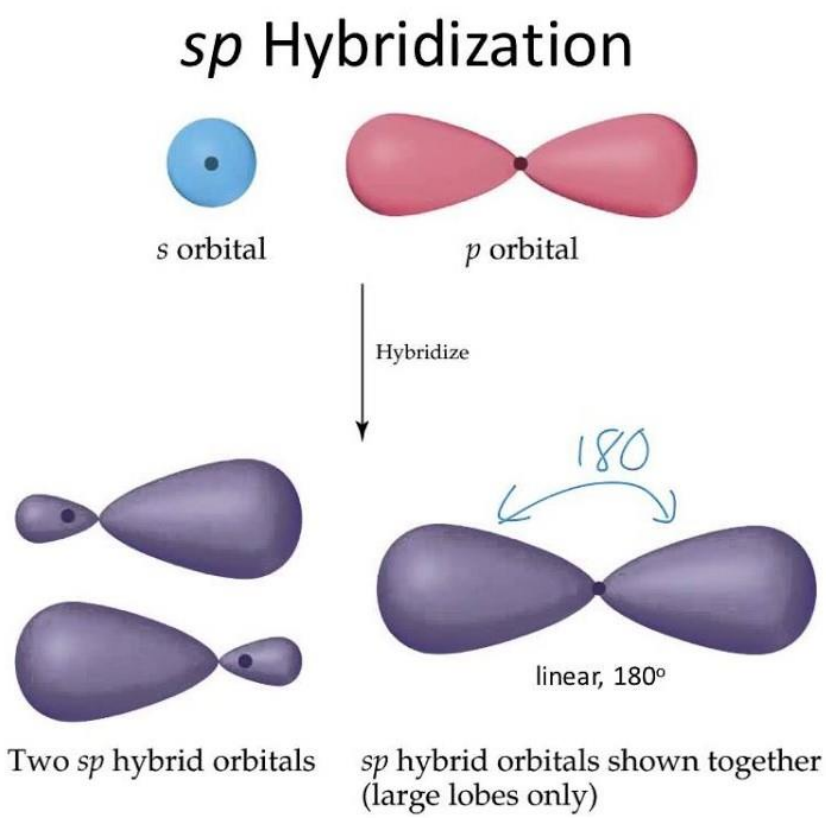


- shared 1*s* electron from 1st hydrogen atom
- shared 1*s* electron from 2nd hydrogen atom

Fig. 2.12, *Callister & Rethwisch 9e.*

Bond Hybrization

Bond hybridization **occurs when an atom bonds using electrons from both the s and p orbitals, creating an imbalance in the energy levels of the electrons**. To equalize these energy levels, the s and p orbitals involved are combined to create hybrid orbital



Bond Hybrization

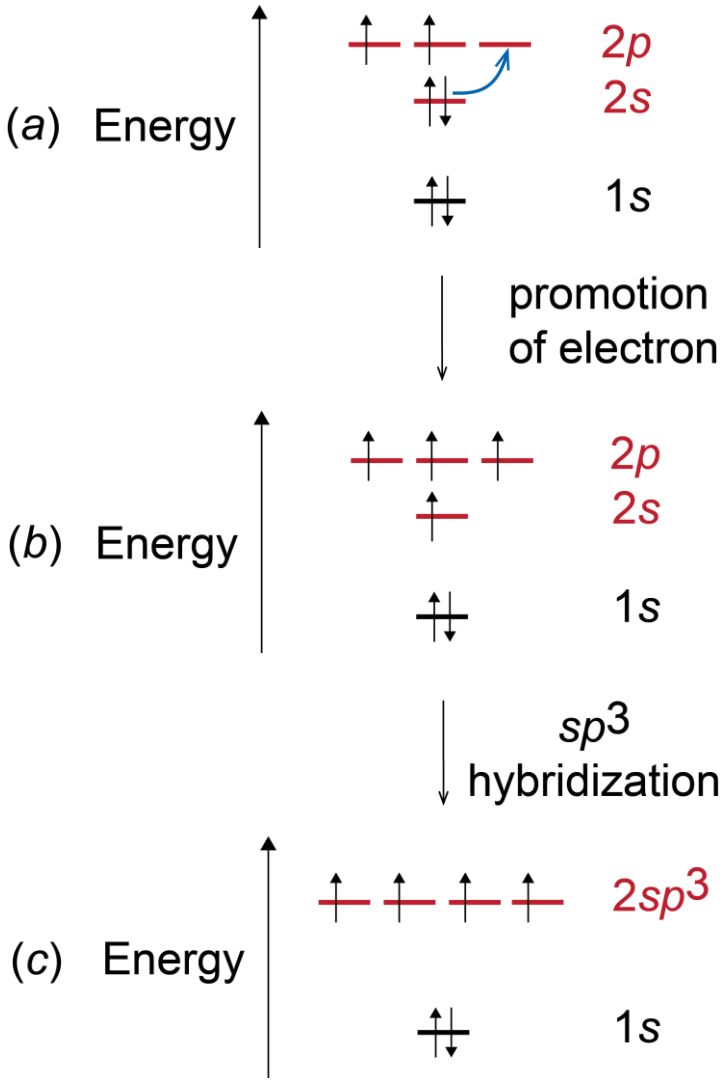


Fig. 2.13, *Callister & Rethwisch 9e.*

- Carbon can form sp^3 hybrid orbitals

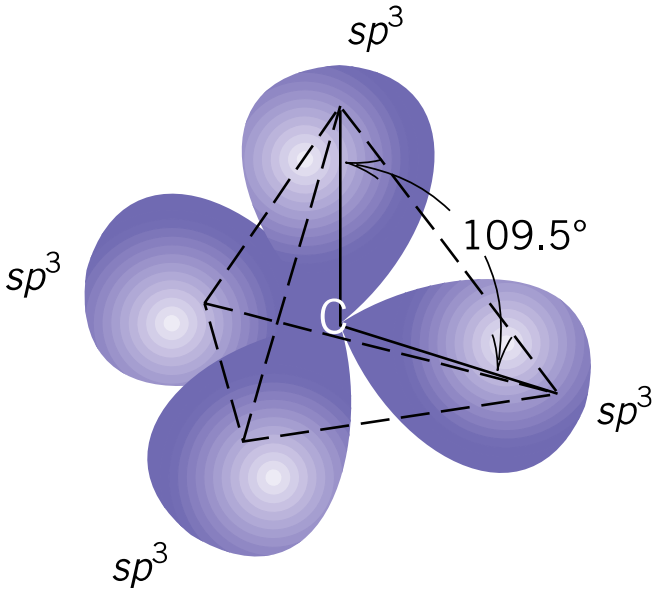


Fig. 2.14, *Callister & Rethwisch 9e.*
(Adapted from J.E. Brady and F. Senese, *Chemistry: Matter and Its Changes*, 4th edition. Reprinted with permission of John Wiley and Sons, Inc.)

Covalent Bonding: Carbon sp^3

- Example: CH_4

C: has 4 valence e^-, needs 4 more

H: has 1 valence e^-, needs 1 more

Electronegativities of C and H are comparable so electrons are shared in covalent bonds.

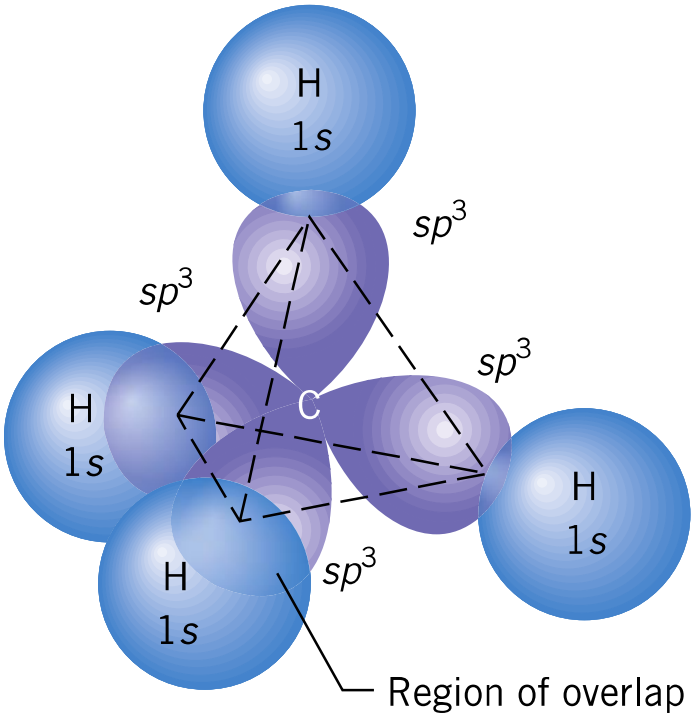


Fig. 2.15, *Callister & Rethwisch 9e.*
(Adapted from J.E. Brady and F. Senese, *Chemistry: Matter and Its Changes*, 4th edition. Reprinted with permission of John Wiley and Sons, Inc.)

Primary Bonding

Primary bonds involve sharing or donating electrons between atoms to form a more stable electron configuration

Metallic Bond - are **forces that holds atoms together in a metallic substance.**

Ionic-Covalent Mixed Bonding

$$\%\ \text{ionic character} = \left(1 - e^{-\frac{(X_A - X_B)^2}{4}}\right) \times (100\%)$$

where X_A & X_B are Pauling electronegativities

Ex: MgO

$X_{Mg} = 1.3$

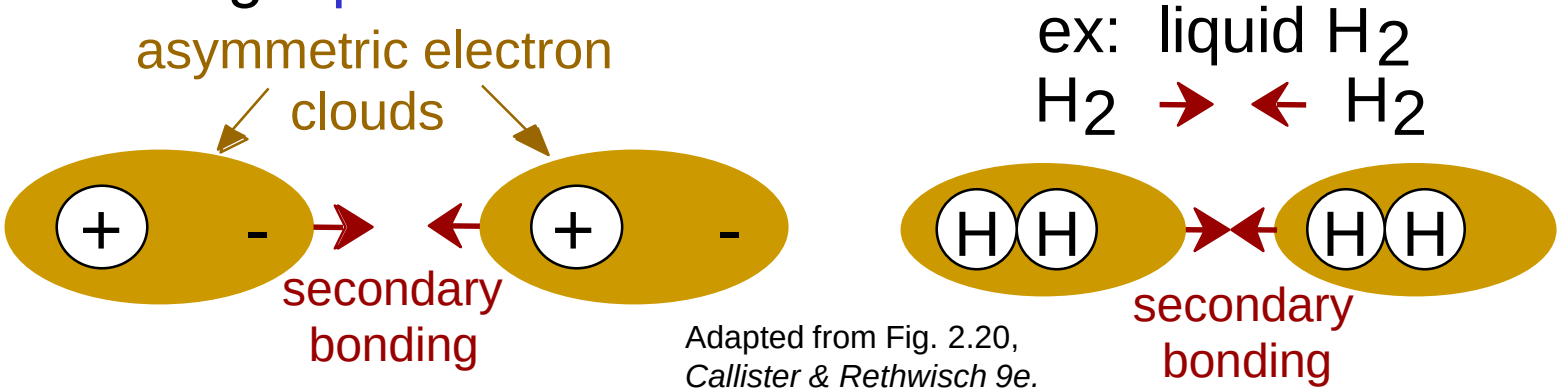
$X_O = 3.5$

$$\%\ \text{ionic character} = \left(1 - e^{-\frac{(3.5-1.3)^2}{4}}\right) \times (100\%) = 70.2\%\ \text{ionic}$$

Secondary Bonding

Arises from interaction between dipoles

- Fluctuating dipoles



Adapted from Fig. 2.20, *Callister & Rethwisch 9e.*

- Permanent dipoles-molecule induced

-general case:

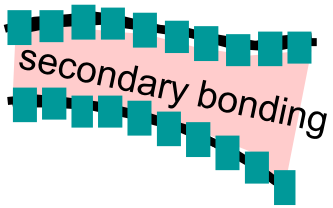


Adapted from Fig. 2.22, *Callister & Rethwisch 9e.*

-ex: liquid HCl



-ex: polymer



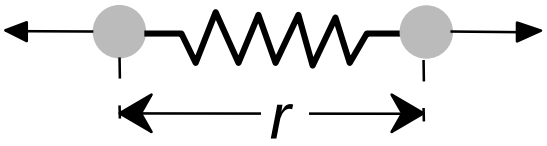
secondary bonding

Summary: Bonding

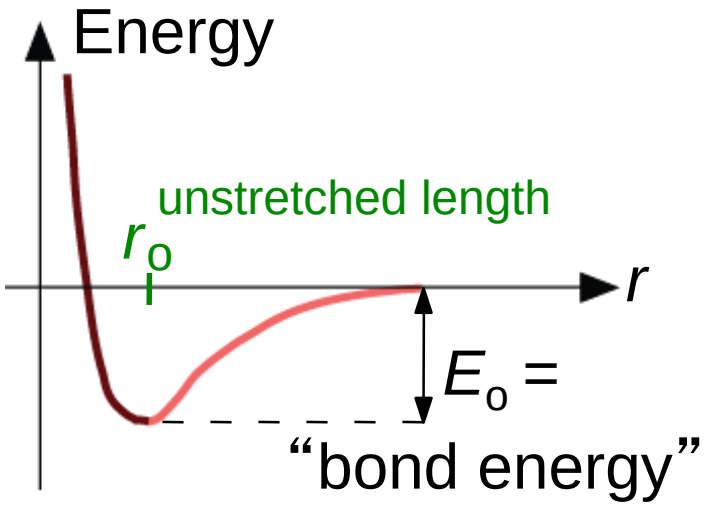
Type	Bond Energy	Comments
Ionic	Large!	Nondirectional (ceramics)
Covalent	Variable large-Diamond small-Bismuth	Directional (semiconductors, ceramics polymer chains)
Metallic	Variable large-Tungsten small-Mercury	Nondirectional (metals)
Secondary	smallest	Directional inter-chain (polymer) inter-molecular

Properties From Bonding: T_m

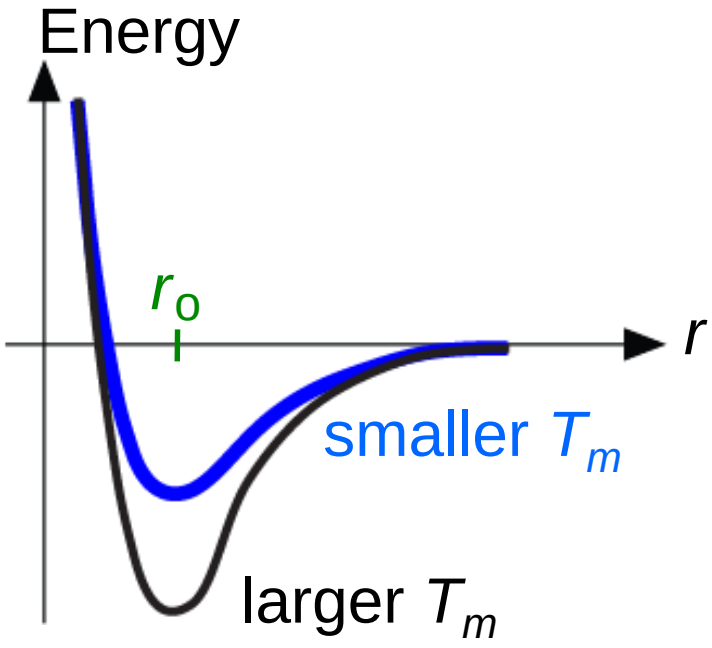
- Bond length, r



- Bond energy, E_o



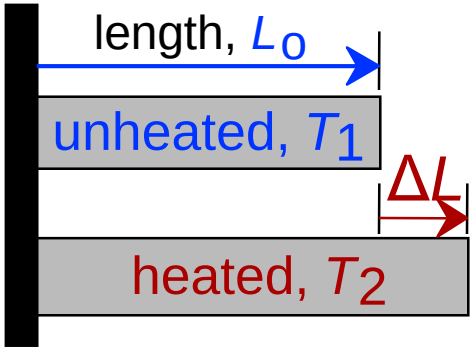
- Melting Temperature, T_m



T_m is larger if E_o is larger.

Properties From Bonding: α

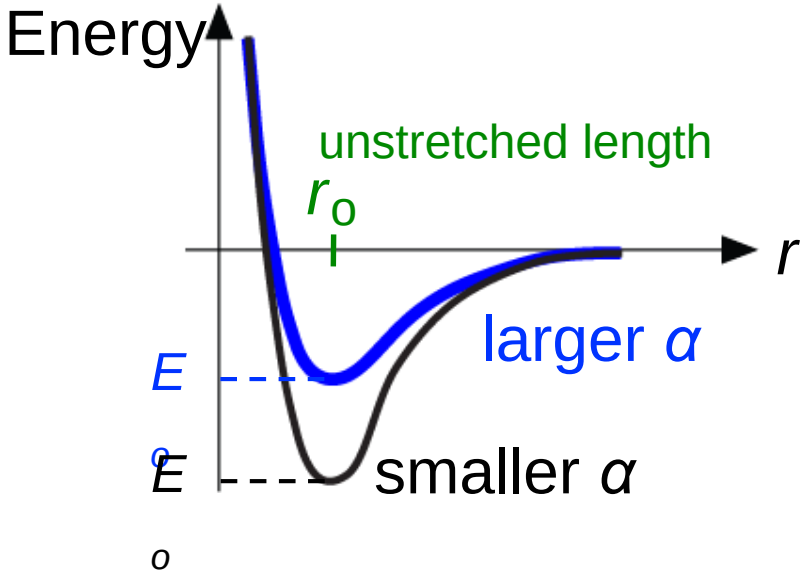
- Coefficient of thermal expansion, α



coeff. thermal expansion

$$\frac{\Delta L}{L_o} = \alpha\,(T_2 - T_1)$$

- α ~ symmetric at r_o



α is larger if E_o is smaller.

Summary: Primary Bonds

Ceramics

(Ionic & covalent bonding):

Large bond energy

large T_m
large E
small α

Metals

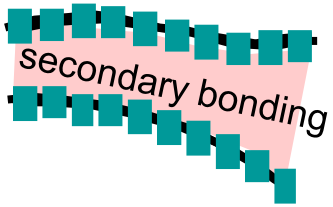
(Metallic bonding):

Variable bond energy

moderate T_m
moderate E
moderate α

Polymers

(Covalent & Secondary):



Directional Properties

Secondary bonding dominates

small T_m
small E
large α

ANNOUNCEMENTS

Reading:

Core Problems:

Self-help Problems: